



KELOMPOK 2

FREDERICK GENOVA VALENCIO

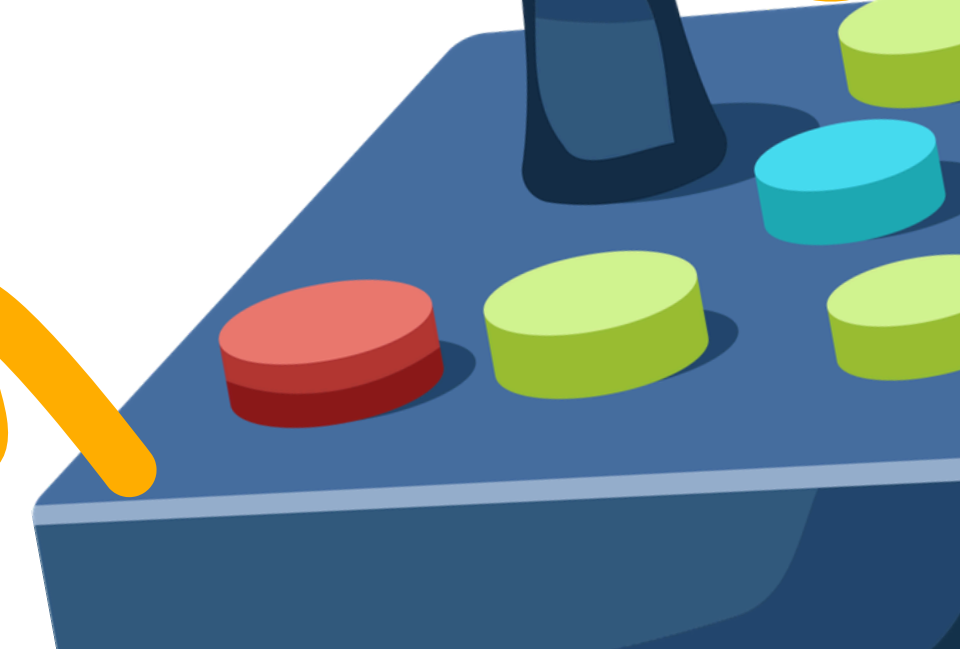
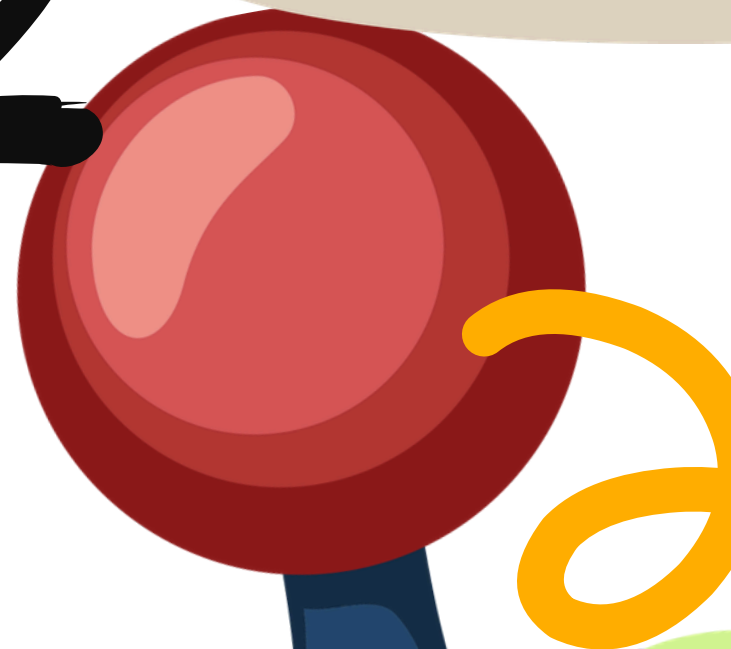
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PROBLEM STATEMENT

The high volume of daily news in today's fast-paced digital era makes it difficult for stakeholders to manually monitor shifts in public perception regarding economic policies. Positive achievements are often obscured by "regulatory noise" due to the high frequency of news coverage, frequently resulting in a fragmented public perception. Furthermore, manual sentiment tracking suffers from strategic blind spots and is highly susceptible to human bias.

To address these challenges, we propose an automated, end-to-end machine learning pipeline utilizing Support Vector Machine (SVM). Implemented as a real-time web application, this technology cuts through dense information overload to offer objective, data-driven insights into the sentiment of financial news in Indonesia.

DATASET SOURCE AND DESCRIPTION

Approximately 60 articles are automatically scraped daily from CNBC Indonesia and Bisnis.com. During this process, the system periodically encountered HTTPConnectionPool timeout issues and network rejections. By implementing robust error-handling and request delays, we successfully bypassed these server restrictions to extract the raw data.

We utilized a hybrid approach for data labeling. A financial lexicon was used to provisionally classify headlines after an initial deduplication process. Crucially, to resolve contextual ambiguities, our team then conducted consensus-based hand annotation. This final balanced dataset yielded a total of 3,300 data rows (1,100 Positive, 1,100 Neutral, and 1,100 Negative).

In line with "Clean Reporting" principles, we identified a labeling artifact. Human annotators occasionally mislabeled macroeconomic reports that were actually neutral and factual, influenced by financial terms that are often politically charged (such as "deficit" or "slump"). Acknowledging this artifact is included to maintain transparency in the downstream model evaluation process.

EXPLORATORY DATA ANALYSIS

The Exploratory Data Analysis (EDA) process was conducted on a dataset comprising 3,300 rows of Indonesian financial news scraped from CNBC Indonesia and Bisnis.com spanning the 2018–2025 period.

This dataset features a perfectly balanced dataset structure, with 1,100 rows allocated to each of the Positive, Neutral, and Negative classes, utilizing the primary textual features from the 'Judul' (Title) column. This balanced distribution was consistently maintained across all experimental stages using a stratified splitting method with an 80% training and 20% testing data split to prevent class bias during model validation.

Beyond mapping class distributions, the analysis also revealed a labeling artifact (human error) driven by highly politicized macroeconomic jargon like "deficit" or "slump." Human annotators occasionally miscategorized objective-neutral news reports as negative—a crucial contextual insight that underscores the necessity of TF-IDF-based feature weighting prior to classification.

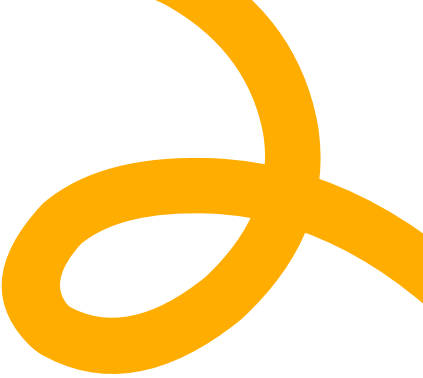
ML PROBLEM FORMULATION

The problem formulation in this project is rooted in the high volume of daily digital financial news reporting, which makes it difficult for stakeholders to manually monitor shifts in public perception regarding economic policies. This manual monitoring has proven to have strategic weaknesses, as it is highly vulnerable to human bias and frequently obscures positive achievements due to elevated regulatory noise stemming from massive news frequencies.

To address these challenges, this problem is formulated into a natural language processing (NLP) machine learning task: automated sentiment classification using the Support Vector Machine (SVM) algorithm. Through an end-to-end system integrated into a real-time web application, this model is designed to cut through information clutter by delivering objective, consistent, and data-driven sentiment analysis of Indonesian financial news.



MODEL SELECTION AND RATIONALE



This experiment establishes Multinomial Naive Bayes as the baseline model. However, it delivered suboptimal performance with an accuracy of 75.45% because its conditional independence assumption analyzes words in isolation. Consequently, it failed to capture crucial contextual nuances, such as the distinction between "laba turun" (declining profits—negative) and "inflasi turun" (declining inflation—positive).

To address this, Support Vector Machine (SVM) was selected due to its exceptional capability in handling the high-dimensional sparse matrices generated by TF-IDF. Instead of analyzing words independently, SVM operates mathematically by determining an optimal hyperplane to distinctly separate the sentiment classes.



Following optimization via GridSearchCV with the hyperparameter parameter $C=1$, the SVM model successfully mitigated the baseline's limitations, significantly boosting performance to reach an accuracy of 93.48% and an F1-Score of 0.9353.



TRAINING AND VALIDATION SETUP

The training and validation data configuration in this project adopts a Hold-Out Validation approach, splitting the balanced 3,300-row financial dataset into a proportion of 80% for training data and 20% for testing data. The data splitting process is strictly implemented using the stratified splitting method to ensure that the distribution of the three sentiment classes (each consisting of 1,100 rows for the Positive, Neutral, and Negative classes) remains proportionally preserved across both the training and testing subsets.

Furthermore, during the experimentation and SVM model tuning phase, the team implemented a 5-Fold Cross-Validation mechanism directly integrated with GridSearchCV, optimized against the weighted F1-score metric. This combination of stratified data partitioning and five-fold cross-validation is systematically designed to reinforce evaluation reliability, mitigate the risk of overfitting, and ensure the model can objectively generalize sentiments from new economic news movements.



DEPLOYMENT ARCHITECTURE



The deployment architecture of this system implements a decoupled client-server architecture pattern to ensure an optimal separation of concerns and high scalability.

- Backend: The AI processing engine built with FastAPI—which loads the Support Vector Machine (SVM) model and handles TF-IDF processing—is implemented as a REST API and deployed on the Hugging Face Spaces platform using Docker containerization.
- Frontend: The interactive user interface is developed using React.js powered by Vite to deliver a responsive and flexible Bring Your Own Device (BYOD) user experience, deployed separately on the Vercel platform.

Data communication between these two independent architectures runs securely and asynchronously via the HTTPS protocol, enabling real-time financial news sentiment analysis without burdening the computing performance on the user's end.

APPLICATION SCREENSHOTS

✦ Coba Langsung Model AI (Real-time)

Ketik judul berita baru, AI kita (SVM) akan menebak sentimennya langsung dari server Python.

Indeks Bisnis-27 Dibuka Melemah, Tertekan Jebloknya Saham BBKA hingga AMRT

Analisis Teks

Hasil Prediksi Model SVM: **Negatif**

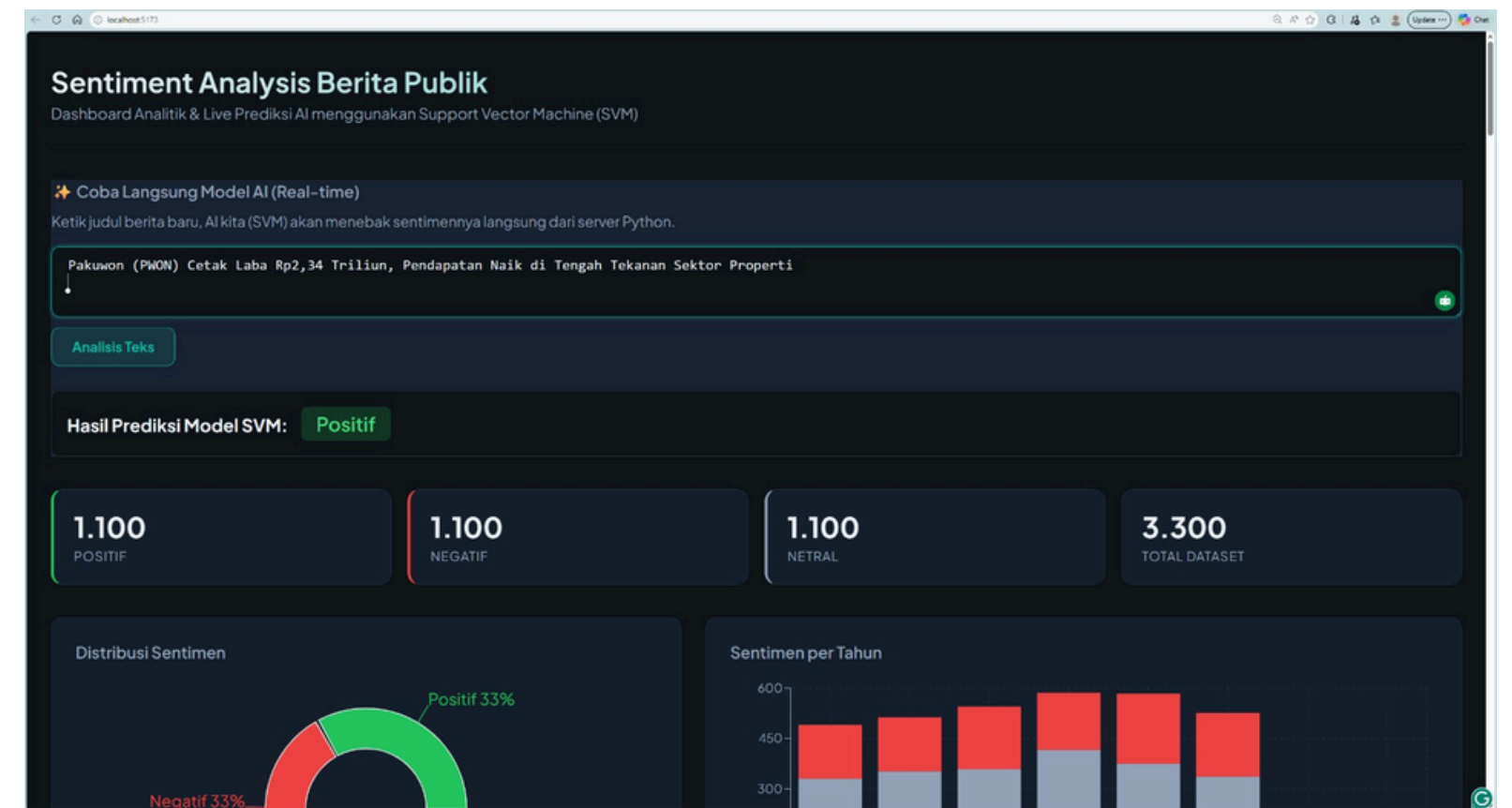
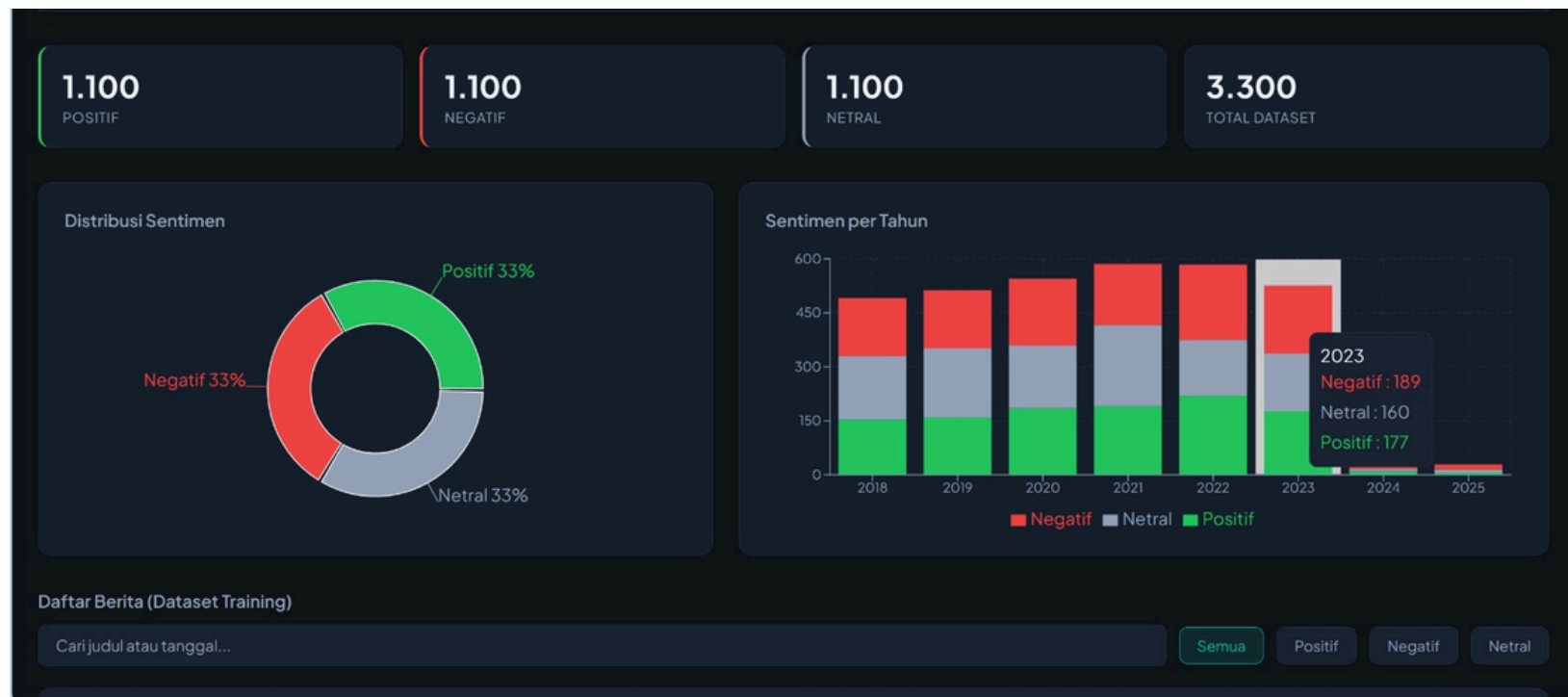
✦ Coba Langsung Model AI (Real-time)

Ketik judul berita baru, AI kita (SVM) akan menebak sentimennya langsung dari server Python.

Emiten Portofolio Lo Kheng Hong (ABMM) Ungkap 2 Fokus Ekspansi pada 2026

Analisis Teks

Hasil Prediksi Model SVM: **Netral**



Daftar Berita (Dataset Training)

19-12-202

Semua Positif Negatif Netral

NO	TANGGAL	JUDUL	SENTIMEN
1	19-12-2020	Uji Coba Sukses, Purwakarta Siapkan Lahan Bawang Merah 30 Hektare	Positif
2	19-12-2020	Libur Nataru, Penumpang di Juanda Diprediksi Mencapai 25.000 Orang	Netral
3	19-12-2020	Liverpool Lagi On Fire, Henderson Ingin Terus Jaga Momentum	Netral
4	19-12-2022	Kasus DBD di Lebak Capai 775 Orang, 4 Dilaporkan Meninggal	Negatif

USER TESTING DESIGN

[Vercel Application Link] → Independent Exploration & Feature Testing → [Google Form Submission]

1. Target Participants

- Profile: Retail Stock Investors & Traders.
- Segmentation: Categorizing responses based on expertise level (Novice / Beginner vs. Experienced).

2. Quantitative Evaluation (Scale 1 - 5)

- UI/UX Usability: Ease of understanding the layout and presentation of the pie charts and bar charts.
- Response Speed: Latency of the AI model in generating real-time sentiment predictions.
- Sentiment Accuracy: Precision of the classification results (Positive/Negative/Neutral) according to user perception.
- Filter Feature Clarity: Functionality and clarity of the search feature within the "News List" table.

3. Qualitative Evaluation & Stress Testing

- AI Stress Testing: Detecting edge cases or vulnerabilities where the AI model "misinterprets" sentence context or sarcasm.
- Utility (Value): Assessing the relevance of data visualization charts in assisting the preliminary analysis of publicly listed companies (issuers).

USER TESTING RESULT

1. User Profile & Segmentation

- Demographics: 80% Gen-Z (18-24 years old) | 20% Millennials (25-34 years old).
- Trading Experience: 60% Intermediate (1-3 years), 20% Beginner, 20% Experienced retail traders.

2. Quantitative Performance (Average Score: 1-5 Scale)

- UI/UX Usability: 4.00 / 5.00 – Analytics charts are intuitive and easy to understand.
- System Latency: 4.00 / 5.00 – Excellent real-time sentiment prediction response.
- Filter Functionality: 3.90 / 5.00 – Search and category filters work effectively.
- Perceived Utility: 3.80 / 5.00 – Highly useful as a quick screening tool for market sentiment.
- Sentiment Accuracy: 3.40 / 5.00 – Primary bottleneck; requires model refinement.

3. AI Stress Test & Vulnerabilities

- Keyword Bias vs. Semantic Context: The SVM model struggles with complex financial syntax. For instance, "penurunan Kerugian sebesar 80%" (an 80% reduction in losses – Positive) was misclassified as Neutral because the model overindexed on the negative Keyword "Kerugian" (loss).
- Geopolitical Edge Cases: Complex sentences mixing political figures with market outlooks trigger false positives.

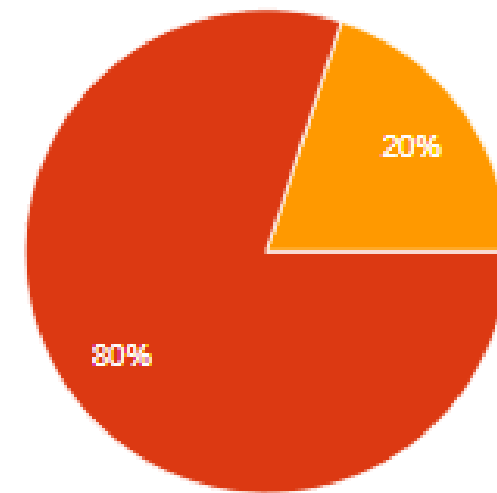
4. Actionable Next Steps

- Model Optimization: Upgrade feature weighting (e.g., use bigrams/trigrams) to better capture negations and financial context.
- Confidence Score: Add a prediction confidence percentage to increase AI transparency for ambiguous headlines.
- UX Improvement: Implement a brief interactive onboarding tour for new users and ensure the dashboard displays the most updated news.

USER TESTING RESULT

Usia:

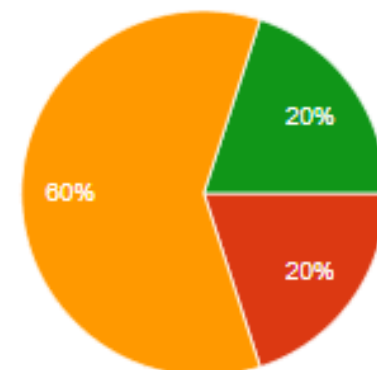
5 responses



- Di bawah 18 tahun
- 18 - 24 tahun
- 25 - 34 tahun
- 35 - 44 tahun
- Di atas 45 tahun

Pengalaman dalam Trading / Investasi Saham:

5 responses



- Belum pernah (Awam)
- Pemula (Kurang dari 1 tahun)
- Menengah (1 - 3 tahun)
- Berpengalaman (Lebih dari 3 tahun)

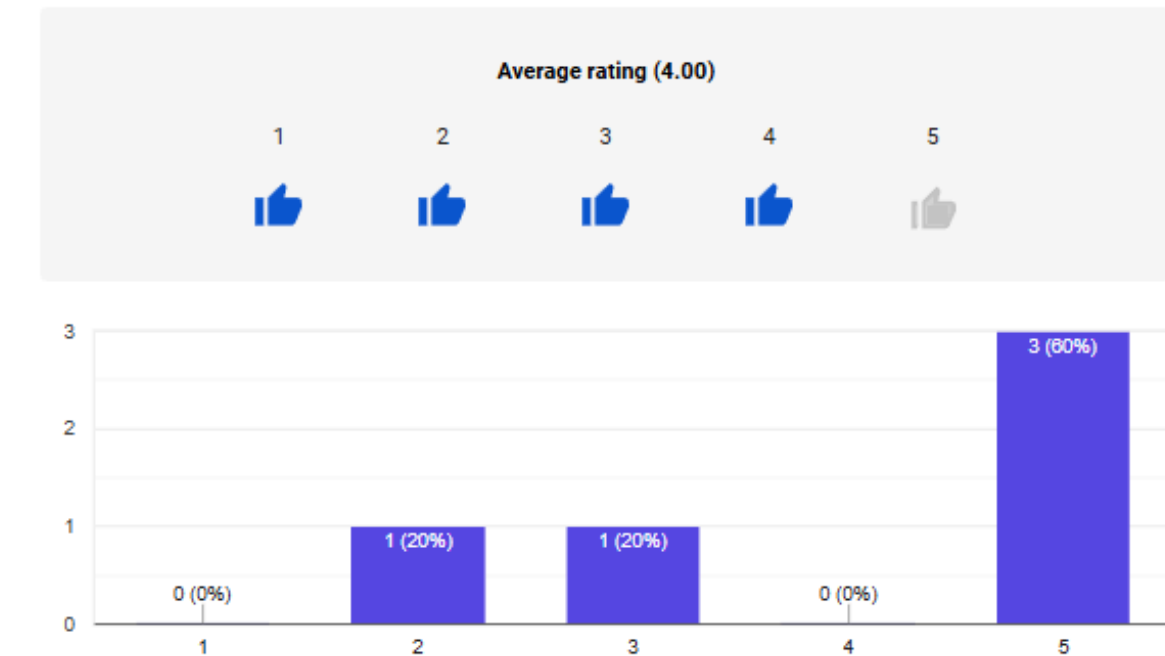
[Copy chart](#)

A. Evaluasi Kuantitatif (Skala 1 - 5)(Keterangan: 1 = Sangat Kurang / Sangat Tidak Setuju, 5 = Sangat Baik / Sangat Setuju)

Kenyamanan Tampilan (UI/UX): Seberapa intuitif dan mudah dipahami tata letak dashboard analitik (seperti grafik pie dan bar chart) pada pandangan pertama?

[Copy chart](#)

5 responses



B. Evaluasi Kualitatif (Feedback Terbuka)

(Mohon berikan jawaban singkat berdasarkan pengalaman Anda mencoba website ini)

Uji Tekanan (Stress Test) AI: Saat Anda mencoba mengetikkan judul berita secara real-time, apakah ada hasil klasifikasi yang menurut Anda "salah tangkap" konteksnya? (Misal: berita yang seharusnya bermakna positif, tapi malah terbaca negatif/netral. Jika ada, sebutkan contohnya).

5 responses

Saya masukan kalimat "moodys menurunkan outlook pasar saham indonesia akibat prabowo dilengserkan" malah positif

sejauh ini tidak ada

sejauh ini belum ada miss data yang saya dapati

Beberapa berita yang saya input masih salah sentimen

Ya, ditemukan kasus salah klasifikasi. Saat diuji dengan judul "Perusahaan mencatat penurunan kerugian sebesar 80% pada kuartal pertama", model SVM mengklasifikasikan sentimen sebagai Netral, padahal secara konteks berita tersebut seharusnya Positif karena menunjukkan perbaikan kinerja perusahaan. Hal ini terjadi karena model lebih banyak mengenali kata kunci seperti kerugian dibanding memahami makna keseluruhan kalimat.

ANALYSIS AND DISCUSSION

Analysis :

- The SVM model achieved 93.48% accuracy and 0.9353 F1-score, outperforming the Multinomial Naive Bayes baseline (75.45% accuracy).
- The Balanced dataset helped the model learn all sentiment classes fairly.
- TF-IDF and SVM effectively captured important patterns in financial news headline.

Discussion :

- Most predictions were accurate for headlines with clear positive or negative sentiment.
- Misclassifications occurred in context-dependent headlines, especially those containing financial terms such as loss, deficit, or inflation.
- User testing showed that the application is easy to use and provides fast realtime predictions.



LIMITATIONS



1. Trapped in Formal Language (False Neutrals)

The overly rigid and objective writing style of economic journalists often masks the news story's true "emotional" undertones.

Consequence: The news story fails to be detected and is mistakenly classified as neutral.

2. "Blind" to Sentence Context (Lexical Bias)

The model sometimes overreacts to a single extreme word (e.g., "crisis," "deficit").

Consequence: Headlines with positive sentiment, such as "Company X Is Crisis-Free," may be incorrectly predicted as negative.

3. Limited to News Headlines

Sentiment accuracy is currently derived solely from the short text of the headline; it does not yet account for the weight of sentences within the full body of the article.

CONCLUSION AND FUTURE WORK

Conclusion :

- A realtime sentiment analysis system for Indonesian financial news was successfully developed using TF-IDF and Support Vector machine (SVM)
- The model achieved 93.48% accuracy and a 0.9353 Weighted F1-Score, significantly outperforming the Multinomial Naive Bayes baseline.
- The Application can classify financial news into Positive, Neutral, and Negative sentiments effectively and provide realtime insights for users.

Future Work:

- Expand the dataset with more recent financial news articles.
- Utilize full news articles instead of headlines only.
- Implement advanced NLP models such as BERT or IndoBERT for better contextual understanding.
- Improve handling of ambiguous and context-dependent financial terminology.
- Enhance the web application with additional analytics and visualization features.



THANK YOU

